Individual Project

-Research on Unit Testing-

Due to time limit, we did not have time to cover 'Testing' topic as the one process of 'Software Engineering' during the semester.

Thus, through this individual project, I want to give you the opportunity to research the topic: "Testing". There are 4 types of the testing technologies: Unit Testing, Integration Testing, System Testing and Acceptance Testing. We want to focus on the 'Unit Testing'.

Here are materials you need to study for this individual project:

1. Video Resources:

1) The Clean Code Talks -- Unit Testing

2) Using JUnit Testing

This course has many modules. Feel free to watch as many as you want to fill-in your knowledge of using JUnit for unit testing Java applications. You are required to watch the following modules:

- Assertions (5:44)
- Basic Unit Tests (5:36)
- Exception Handling (2:46)

These videos were done before the release of JUnit 5. JUnit 5 has added many features that you will probably not use. There are some notable differences with what you will see in these videos and JUnit 5 unit tests. These include:

- All the assertion statements can take a message but it is the last parameter in the assertion call rather than the first.
- You use an `assertThrows` assertion for testing that an exception is thrown rather than specifying this on the `@Test` line before the test method begins. `assertThrows` is given the exception class that is expected to be thrown, and an executable that should throw the exception.

2. Web Articles and Blogs:

- Martin Fowler
    - Unit Test
    - Mocks aren't stubs
    - Test-Driven Development
- What-makes-a-good-unit-test; read the approved answer
- Key qualities of a good unit test
- 5 Questions Every Unit Test Must Answer How to Write Better Tests
- Unit vs Functional vs Integration Tests

3. Technical Resources:

- JUnit
- Mockito

4. Wikipeida

- Software testing
- Software testability
- Unit testing
- Mock object

After studying the above 4 categories, **you need to prepare for a presentation**. The presentation should include all the main components of materials in each catetory. The duration of the presentation should be at least **30 minutes**. The presentation should be recorded and uploaded on Youtube.

You also need to create **ppt slides** for the presentation.

In sum, please submit the ppt slides which include the link to your video presentation by 11:59pm, December 2.

The class on **November 25 (Mon)** will be used to prepare for this presentation. So, we do not meet on that day (11/25). Just prepare the presentation and upload your video on YouTube.

This will be a part of the final exam.

So, if you do not submit the presentation, you cannot pass the course: Software Engineering.